

Combined Oxidative Phosphorylation Deficiency 51 (COXPD51): A Comprehensive Disease Characteristics Report

MONDO ID: MONDO:0033631 | **OMIM:** #619057 | **Gene:** *PTCD3* (*MRPS39*), OMIM *614918 | **Category:** Mendelian, autosomal recessive

Summary

Combined Oxidative Phosphorylation Deficiency 51 (COXPD51) is an ultra-rare, autosomal-recessive primary mitochondrial disorder caused by **biallelic loss-of-function variants in *PTCD3*** (also called *MRPS39*), a nuclear gene on chromosome 2p11.2 that encodes the **small mitoribosomal subunit protein mS39**. mS39 is an RNA-binding pentatricopeptide-repeat (PPR) protein and an essential structural/functional component of the 28S small subunit of the mitochondrial ribosome. When both alleles are disrupted, assembly and function of the small mitoribosomal subunit fail, mitochondrial mRNA translation is broadly impaired, and the 13 mtDNA-encoded OXPHOS subunits are not synthesized in adequate quantity. The downstream result is a **combined deficiency of respiratory-chain complexes I and IV (and, in some experimental systems, complex III)**, collapse of oxidative ATP production, and — because neurons and other high-energy-demand tissues are exquisitely dependent on OXPHOS — a severe neurodegenerative phenotype.

Clinically, COXPD51 manifests as **early-infantile Leigh syndrome** (subacute necrotizing encephalomyelopathy). Reported patients present in the first months of life with psychomotor delay and regression, dystonia, optic atrophy, nystagmus, tonic-clonic seizures, respiratory insufficiency, and feeding difficulties. Brain MRI shows the bilateral, symmetric signal abnormalities of the basal ganglia and brainstem, together with thalamic changes and optic-

nerve atrophy, that define the Leigh pattern. The disease is severe and progressive with a poor prognosis; treatment is entirely supportive, as there is no disease-modifying or curative therapy.

The disease is extraordinarily rare: as of this report, only **four molecularly-confirmed patients** have been fully published (one by Bornha et al. 2019; three by Muñoz-Pujol et al. 2023). The genetic mechanism, biochemical consequences, and clinical spectrum are, however, unusually well-defined for such a rare entity thanks to functional rescue (complementation) experiments in patient fibroblasts, siRNA knockdown studies in cell lines, and — importantly for gene essentiality — an International Mouse Phenotyping Consortium (IMPC) knockout showing that homozygous *Ptcd3* deletion is embryonic-lethal. This last point explains why living patients only ever carry **hypomorphic** allele combinations rather than complete-null genotypes.

Key Findings

Finding 1 — COXPD51 is caused by biallelic loss-of-function variants in *PTCD3* (*MRPS39*)

The genetic cause of COXPD51 is firmly established as **recessive, biallelic loss of function in *PTCD3***. The disease was first delineated by Bornha et al. (2019), who used exome sequencing in a single patient to identify two loss-of-function variants: a canonical splice-acceptor change **c.415-2A>G** and a frameshift insertion **c.1747_1748insCT (p.Phe583Serfs*3)**. As the authors state, *"Exome sequencing revealed two potentially loss-of-function variants [c.415-2A>G, and c.1747_1748insCT (p.Phe583Serfs*3)] in PTCD3 (also known as MRPS39). PTCD3, a member of the pentatricopeptide repeat domain protein family, is a component of the small mitochondrial subunit"* (PMID: 30607703).

The gene–disease relationship was independently confirmed and elevated to a definitive association by Muñoz-Pujol et al. (2023), who reported three additional patients from two families carrying compound-heterozygous variants — *"WES and RNA-seq identified compound heterozygous variants in PTCD3 in both families: c.[1453-1G>C];[1918C>G] and c.[710del];[902C>T]"* (PMID: 36450274). These families demonstrated reduced PTCD3 protein and severe reductions in complex I and complex IV subunit steady-state levels and activities, confirming pathogenicity. Two independent reports, in unrelated families, satisfy standard gene-validity criteria for a definitive Mendelian gene–disease relationship.

Finding 2 — COXPD51 manifests clinically as early-infantile Leigh syndrome with optic atrophy

Across all four reported patients, the clinical picture is that of **Leigh syndrome** with onset in the first months of life. Core features include psychomotor delay/regression, respiratory insufficiency, and feeding difficulties, with a neurologic phenotype of **dystonia, optic atrophy, nystagmus, and tonic-clonic seizures**. Muñoz-Pujol et al. summarize: *"The patients presented in the first months of life with psychomotor delay, respiratory insufficiency and feeding difficulties. The neurologic phenotype included dystonia, optic atrophy, nystagmus and tonic-clonic seizures. Brain MRI showed optic nerve atrophy and thalamic changes, consistent with Leigh syndrome"* (PMID: 36450274). Their paper title makes the conclusion explicit: *"Leigh syndrome is the main clinical characteristic of PTC3 deficiency."*

The index patient of Borna et al. showed a concordant picture: *"We describe a patient who presented with low birth weight, mental retardation, and optic atrophy. Brain MRI showed abnormal bilateral signals at the basal ganglia and brainstem, and the patient was diagnosed as Leigh syndrome"* (PMID: 30607703).

Suggested HPO terms: Leigh-like disease / abnormality of the basal ganglia (HP:0002134), Optic atrophy (HP:0000648), Nystagmus (HP:0000639), Dystonia (HP:0001332), Generalized tonic-clonic seizures (HP:0002069), Global developmental delay (HP:0001263), Developmental regression (HP:0002376), Feeding difficulties (HP:0011968), Respiratory insufficiency (HP:0002093), Elevated circulating lactate concentration (HP:0002151), Intrauterine growth retardation / low birth weight (HP:0001518).

Finding 3 — Molecular mechanism: mS39 loss impairs mitochondrial translation, causing combined complex I + IV deficiency

The mechanistic chain is well supported. UniProt entry **Q96EY7** defines PTC3 as *"Small ribosomal subunit protein mS39"* (689 aa), a mitochondrial RNA-binding protein and component of the mitochondrial small ribosomal subunit (28S mt-SSU) associated with the 12S mt-rRNA. Relevant GO terms include **GO:0005763 (mitochondrial small ribosomal subunit)**, **GO:0032543 (mitochondrial translation)**, and **GO:0019843 (rRNA binding)**.

The functional role was first established by Davies et al. (2009), who showed that *"lowering PTC3 in 143B osteosarcoma cells decreased mitochondrial protein synthesis, mitochondrial respiration and the activity of Complexes III and IV, suggesting that PTC3 has a role in mitochondrial translation"* (PMID: 19427859).

In patient cells, Borna et al. demonstrated the complete causal cascade: *"The patient had marked decreases in mitochondrial complex I and IV levels and activities, oxygen consumption and ATP biosynthesis, and generalized mitochondrial translation defects in fibroblasts. Quantitative proteomic analysis revealed decreased levels of the small mitoribosomal subunits"* (PMID: 30607703). Critically, they proved causation by rescue: *"Complementation experiments rescued oxidative phosphorylation complex I and IV levels and activities, ATP biosynthesis, and MT-RNR1 rRNA transcript level, providing functional validation."* Re-introduction of wild-type PTC3 restored the biochemical defect, closing the loop between genotype and phenotype. Muñoz-Pujol et al. reproduced this in additional families with reduced PTC3 protein and severe reductions in complex I and IV subunit steady-state levels/activities and respiration.

Finding 4 — Variant spectrum: autosomal-recessive, biallelic, ultra-rare loss-of-function alleles

PTCD3/*MRPS39* is located at **chr2p11.2** (GRCh38 chr2:86,106,223–86,142,157), **HGNC:24717**, **OMIM *614918**, **Ensembl ENSG00000132300**, RefSeq **NM_017952.6**. gnomAD constraint metrics indicate that heterozygous loss of function is tolerated (pLI $\approx$ 0; LOEUF/oe_lof $\approx$ 0.87; observed 89 / expected 102 LoF alleles), which is fully consistent with a **recessive** disease mechanism — carriers are unaffected.

ClinVar lists 223 *PTCD3* records (17 Pathogenic, 10 Likely pathogenic, 129 VUS, plus benign/conflicting). The pathogenic/likely-pathogenic variants are predominantly loss-of-function:

Variant class	Examples
Frameshift	c.695del; c.710del (p.Thr237fs); c.1746_1747dup (p.Phe583fs); c.1747_1748insCT (p.Phe583Serfs*3)
Nonsense	c.640C>T (p.Gln214*); c.1166C>G (p.Ser389*); c.1431G>A (p.Trp477*)
Canonical splice-site	c.415-2A>G; c.805-2A>G; c.1148-2A>G; c.1453-1G>C; c.1630-1G>A; c.1979+1G>A
Missense (recurrent)	c.902C>T (p.Thr301Ile); c.1918C>G (p.Pro640Ala)
Structural	large 2p11.2 deletion encompassing <i>PTCD3</i>

Population allele frequencies (gnomAD v4) confirm that disease alleles are **ultra-rare or absent** (e.g., c.902C>T AF $\approx$ 3.4×10^{-6} ; c.1453-1G>C AF $\approx$ 2.1×10^{-6} ; c.415-2A>G AF $\approx$ 6.2×10^{-6} ; c.710del absent). The recurrent missense **c.1918C>G (p.Pro640Ala)** is comparatively common

(AF $\approx$ 0.0014) and ClinVar-conflicting, consistent with a **hypomorphic allele** that produces disease only when inherited *in trans* with a severe (null-like) allele. This aligns with the two source publications (PMID: 30607703; PMID: 36450274) showing splice/frameshift alleles combined with missense changes. All origins are **germline**; no somatic mechanism applies to this disease.

Finding 5 — Prognosis is poor: infantile-onset Leigh syndrome with early mortality

Because only four molecularly-confirmed patients are published, dedicated COXPD51 natural-history data do not exist; the best prognostic proxy is the natural history of **Leigh syndrome** as a whole, which is severe and often fatal in early childhood, especially with early onset. Ogawa et al. (2020), studying 166 Japanese Leigh patients, reported that *"Nearly 90% of deaths occurred by age 6. Mortality rate of patients with onset before 6 months of age was significantly higher than that of onset after 6 months. All patients with neonatal onset were either deceased or bedridden"* (PMID: 31967322). Overall mortality in that cohort was 24.1%.

Because *PTCD3* is a **nuclear** gene, COXPD51 falls into the nuclear-DNA (nDNA) Leigh category, which tends to present earlier than mtDNA-caused disease. The meta-analysis by Chang et al. (2020) found *"Patients with nDNA mutations were younger than those with mtDNA mutations (8.82 ± 13.88 vs 26.20 ± 41.11 years, $P = .007$)"* (PMID: 32000367), and reported the following frequencies across 385 Leigh patients: elevated lactate 72%, developmental retardation 57%, hypotonia 42%, respiratory dysfunction 34%, seizures 33%, poor feeding 29%. Together these support an early-onset, severe, progressive course for COXPD51.

Finding 6 — Model systems: in vitro / patient-cell models exist; no dedicated animal disease model

Functional understanding of COXPD51 derives principally from **cellular models**: (i) siRNA knockdown of *PTCD3* in 143B osteosarcoma cells (Davies 2009), which reduced mitochondrial protein synthesis, respiration, and complex III/IV activity; and (ii) **patient-derived dermal fibroblasts** (Borna 2019; Muñoz-Pujol 2023), which recapitulate the biochemical defect (combined complex I + IV deficiency, translation defect) and were used for lentiviral complementation/rescue. Conserved orthologs are annotated in NCBI Gene: **mouse** *Ptcd3* (GeneID 69956), **rat** *Ptcd3* (500199), **zebrafish** *ptcd3* (553325); the budding-yeast ortholog of mS39 is **RSM22**. No published knockout/knock-in mouse, zebrafish, or invertebrate line has been specifically phenotyped as a COXPD51 *disease* model.

Finding 7 — IMPC *Ptcd3* knockout: homozygous embryonic lethality confirms gene essentiality

The International Mouse Phenotyping Consortium (IMPC) genotype–phenotype records for *Ptcd3* provide a decisive insight into gene essentiality. Homozygous knockout produces **"preweaning lethality, complete penetrance"** and **"embryonic lethality prior to organogenesis."** Heterozygous animals show subtle phenotypes: **impaired pupillary reflex** (pupil light response), decreased circulating HDL cholesterol, and increased mean corpuscular hemoglobin. The embryonic lethality of the complete null explains a key genetic constraint of COXPD51: **surviving patients cannot be complete nulls** — they must retain some residual mS39 function, which is why all reported genotypes pair a severe allele with a hypomorphic (often missense/splice) allele that preserves partial activity.

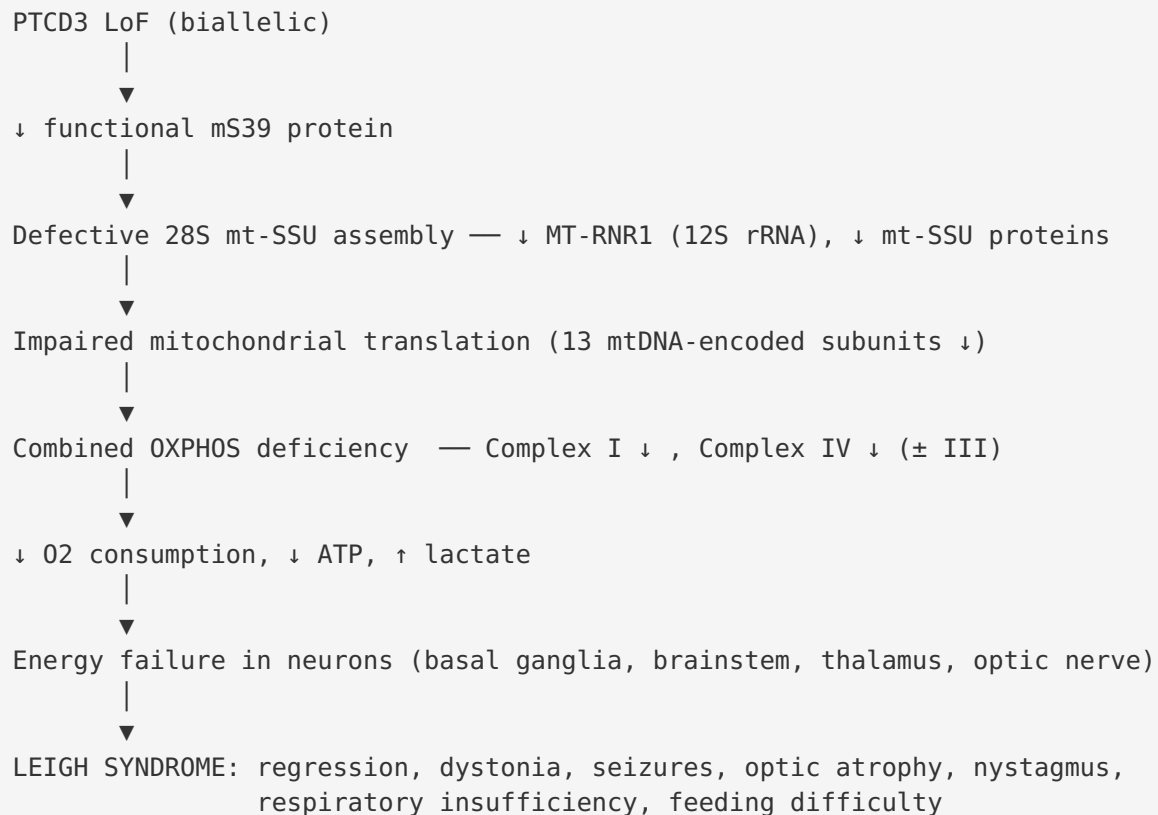
Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in *PTCD3*** (splice-site, frameshift, nonsense, or hypomorphic missense in *trans*) **lead to** reduced or dysfunctional mS39 protein.
2. Reduced functional mS39 **results in** impaired assembly and stability of the **28S small mitoribosomal subunit** (mt-SSU); quantitative proteomics shows decreased small mitoribosomal subunit proteins and reduced 12S rRNA (MT-RNR1). (*Demonstrated in patient fibroblasts.*)
3. Defective mt-SSU **leads to a generalized defect in mitochondrial mRNA translation** — the 13 mtDNA-encoded OXPHOS subunits are undersynthesized. (*Demonstrated.*)
4. Reduced mtDNA-encoded subunit synthesis **results in** failed assembly of respiratory-chain complexes, producing a **combined deficiency of complex I and complex IV** (± complex III in knockdown systems), since these complexes contain mtDNA-encoded core subunits. (*Demonstrated biochemically; rescued by complementation.*)
5. Combined OXPHOS deficiency **leads to** decreased oxygen consumption and **collapse of oxidative ATP synthesis**, with a compensatory shift to glycolysis and **lactic acidosis**. (*Demonstrated.*)
6. Energy failure **results in** injury to high-energy-demand post-mitotic tissues — chiefly **neurons of the basal ganglia, brainstem, thalamus, and optic pathway** — producing the

bilateral symmetric necrotizing lesions characteristic of Leigh syndrome. (*Inferred by analogy to Leigh-syndrome pathophysiology; supported by MRI pattern.*)

7. Regional neurodegeneration **leads to** the clinical phenotype: psychomotor regression, dystonia, seizures, optic atrophy, nystagmus, respiratory insufficiency, and feeding difficulty. (*Demonstrated clinically.*)



Anatomical, cellular, and subcellular mapping

Level	Structures involved	Suggested ontology terms
Subcellular	Mitochondrion; mitochondrial small ribosomal subunit; matrix	GO:0005739 (mitochondrion); GO:0005763 (mitochondrial small ribosomal subunit); GO:0005759 (mitochondrial matrix)
Biological process	Mitochondrial translation; rRNA binding; OXPHOS / ATP synthesis	GO:0032543 (mitochondrial translation); GO:0019843 (rRNA binding); GO:0006119 (oxidative phosphorylation)
Cell type	Neurons (esp. basal ganglia / brainstem / retinal ganglion cells of optic nerve)	CL:0000540 (neuron); CL:0000099 (interneuron); CL:0000740 (retinal ganglion cell)
Tissue	Nervous tissue; gray-matter nuclei	UBERON:0001021 (nerve); UBERON:0002020 (gray matter)
Organ / site	Basal ganglia, brainstem, thalamus, optic nerve	UBERON:0002420 (basal ganglion); UBERON:0002298 (brainstem); UBERON:0001897 (thalamus); UBERON:0000941 (optic nerve)
Body system	Central nervous system (primary); respiratory drive (secondary)	UBERON:0001017 (central nervous system)

Chemical entities (CHEBI): ATP (CHEBI:15422), lactate/lactic acid (CHEBI:24996), oxygen (CHEBI:15379).

Why complexes I and IV specifically?

Complexes I, III, IV, and V all contain mtDNA-encoded core subunits and therefore depend on mitochondrial translation; **complex II is entirely nuclear-encoded and is spared**. Complex I (7 mtDNA subunits) and complex IV (3 mtDNA subunits) are the most sensitive readouts of a global mitochondrial translation lesion, which is why the biochemical signature of COXPD51 is a **combined complex I + IV deficiency** with preserved complex II — a fingerprint shared across the mitochondrial-translation / mitoribosomopathy class of disorders (e.g., COXPD7/*C12orf65*, COXPD9/*MRPL3*).

Evidence Base

PMID	Title / focus	Evidence type	Contribution
30607703	<i>Mitochondrial ribosomal protein PTC3 mutations cause oxidative phosphorylation defects with Leigh syndrome</i> (Borna et al. 2019)	Human clinical + in vitro rescue	First COXPD51 patient; identifies biallelic LoF variants; demonstrates translation defect, combined CI/CIV deficiency; complementation rescue = functional validation
36450274	<i>Leigh syndrome is the main clinical characteristic of PTC3 deficiency</i> (Muñoz-Pujol et al. 2023)	Human clinical	3 additional patients / 2 families; compound-het variants; establishes Leigh syndrome as the defining phenotype; upgrades gene–disease validity
19427859	<i>Pentatricopeptide repeat domain protein 3 associates with the mitochondrial small ribosomal subunit and regulates translation</i> (Davies et al. 2009)	In vitro (siRNA)	Defines PTC3 function in mitochondrial translation; knockdown reduces protein synthesis, respiration, and complex III/IV activity
31967322	<i>Mortality of Japanese patients with Leigh syndrome</i> (Ogawa et al. 2020)	Human natural history (n=166)	Prognostic proxy: ~90% of deaths by age 6; early onset = higher mortality; neonatal onset uniformly severe
32000367	<i>A meta-analysis and systematic review of Leigh syndrome</i> (Chang et al. 2020)	Human meta-analysis (n=385)	Phenotype frequencies; nDNA cases have earlier onset than mtDNA cases — supports early severe COXPD51 course
IMPC (Ptc3)	International Mouse Phenotyping Consortium genotype–phenotype records	Model organism	Homozygous KO embryonic-lethal (gene essentiality); heterozygotes show impaired pupillary reflex, altered HDL/MCH

Supporting context on the disease class (mitoribosomopathies / combined OXPHOS deficiencies): COXPD9 due to *MRPL3* (PMID: 34008913) and COXPD7 due to *C12orf65* (PMID: 40993840) present with overlapping Leigh-syndrome / mitochondrial-translation-defect phenotypes, reinforcing the mechanistic placement of COXPD51.

Section-by-Section Synthesis

1. Disease Information

COXPD51 (MONDO:0033631, OMIM #619057) is a Mendelian mitochondrial disease. Synonyms/alternative descriptors: "PTCD3 deficiency," "MRPS39 deficiency," "mitochondrial ribosomal protein mS39 deficiency," and (descriptively) "PTCD3-related Leigh syndrome." No ICD-10 code is specific to COXPD51; it maps to broad mitochondrial-disease/Leigh categories (ICD-10 G31.8 / E88.40 range; ICD-11 8C73/5C53 area). No specific Orphanet number is assigned to the PTCD3 subtype; it falls under the Leigh syndrome / combined OXPHOS deficiency umbrella. Information is derived from **aggregated disease-level resources and individual case reports** (four molecularly-confirmed patients), not EHR/registry data.

2. Etiology

Cause: purely genetic — biallelic (recessive) loss-of-function variants in *PTCD3*. **Genetic risk factor:** carrier status in both parents; consanguinity/founder effects are plausible but not established given the tiny cohort. **Environmental risk/protective factors:** none identified; this is a monogenic disorder. **Gene–environment interactions:** not applicable / not reported. The hypomorphic p.Pro640Ala missense allele acts as a genetic modifier that permits survival when paired with a severe allele.

3. Phenotypes

See Finding 2 and HPO terms above. Age of onset: **neonatal to early infancy** (first months of life). Severity: **severe**. Progression: **progressive with regression**. Frequency across the 4 patients: optic atrophy, developmental delay/regression, and Leigh-pattern MRI are near-universal; dystonia, nystagmus, seizures, respiratory insufficiency, and feeding difficulty are common. Quality-of-life impact is profound (severe disability; dependence for all care).

4. Genetic / Molecular Information

See Finding 4. Causal gene *PTCD3/MRPS39* (HGNC:24717, OMIM *614918, chr2p11.2). Variant classes: splice, frameshift, nonsense, hypomorphic missense, and at least one structural deletion. Inheritance: autosomal recessive. Functional consequence: **loss of function**. Modifier: hypomorphic p.Pro640Ala allele. No epigenetic mechanism is implicated in disease causation (note: unrelated cancer literature describes epigenetic *up*-regulation of PTCD3 in colorectal cancer, [PMID: 40304977](#) — not relevant to COXPD51 pathogenesis).

5. Environmental Information

Not applicable — no environmental, lifestyle, or infectious contributors. Monogenic recessive disease.

6. Mechanism / Pathophysiology

See the ordered causal chain and mechanistic model above.

7. Anatomical Structures Affected

Primary: central nervous system (basal ganglia, brainstem, thalamus, optic nerve). Subcellular: mitochondrion / small mitoribosomal subunit. Lateralization: **bilateral and symmetric** (hallmark of Leigh syndrome). Secondary involvement: respiratory (central hypoventilation) and feeding/growth (failure to thrive).

8. Temporal Development

Onset: **congenital/early-infantile**, insidious-to-subacute. Course: **progressive** neurodegeneration, often punctuated by metabolic decompensation. Duration: chronic but frequently fatal in early childhood (by Leigh-syndrome analogy). Critical period: infancy is the window of maximal vulnerability.

9. Inheritance and Population

Inheritance: **autosomal recessive**; carriers unaffected (heterozygous LoF tolerated per gnomAD). Penetrance: presumed complete for biallelic pathogenic genotypes. Expressivity: variable within a narrow severe range. Prevalence/incidence: **not established** — **ultra-rare** (fewer than ~10 patients worldwide). No confirmed founder effect, sex bias, or ethnic clustering given the small cohort. Carrier frequency: unknown but very low (disease alleles ultra-rare in gnomAD).

10. Diagnostics

- **Biochemistry:** elevated blood/CSF lactate; combined complex I + IV deficiency (with preserved complex II) on respiratory-chain enzymology in muscle/fibroblasts; reduced mitochondrial protein synthesis (in vitro).
- **Imaging:** MRI showing bilateral symmetric T2/FLAIR signal abnormalities of basal ganglia and brainstem, thalamic changes, and optic-nerve atrophy; MR spectroscopy may show a lactate peak.
- **Genetics (definitive): whole-exome or whole-genome sequencing** is the diagnostic method of choice; RNA-seq is useful to resolve splice variants (as in Muñoz-Pujol 2023). Gene-panel testing for mitochondrial/Leigh-syndrome nuclear genes should include *PTCD3*. Single-gene testing is impractical given locus heterogeneity.
- **Differential diagnosis:** other nuclear- and mtDNA-encoded Leigh syndromes and combined-OXPHOS-deficiency mitoribosomopathies (e.g., *NDUFS4*, *SURF1*, *MT-ATP6/T8993C*, *MRPL3* [COXPD9], *C12orf65* [COXPD7]).
- **Screening:** no newborn screening exists; **cascade carrier testing** of relatives and prenatal/preimplantation testing are options once familial variants are known.

11. Outcome / Prognosis

Poor. No curative treatment. Prognosis is inferred from Leigh-syndrome natural history: high early-childhood mortality, especially with onset before 6 months (PMID: 31967322). Morbidity is severe (profound neurodevelopmental disability, dependence). Prognostic factors: age of onset, severity of respiratory involvement, and residual mS39 function conferred by the milder allele.

12. Treatment

No disease-modifying therapy exists. Management is **supportive/symptomatic**, following general mitochondrial-disease and Leigh-syndrome principles: seizure control (anti-seizure medications, avoiding valproate where feasible due to mitochondrial toxicity), nutritional support (gastrostomy for feeding difficulty), respiratory support, physical/occupational therapy, and management of dystonia. "Mitochondrial cocktails" (coenzyme Q10, riboflavin, thiamine, L-carnitine) are commonly used empirically but lack proven efficacy in this specific disorder. No gene, cell, or RNA therapy is available or in trials for COXPD51. **NCIT-relevant supportive categories:** anticonvulsant therapy, nutritional support (NCIT:C15417), physical therapy, respiratory support/supportive care.

13. Prevention

Primary prevention is limited to **genetic counseling and reproductive options** (carrier testing, prenatal diagnosis, preimplantation genetic testing) for at-risk families. No population-level screening, immunization, behavioral, or environmental intervention applies. Tertiary prevention = anticipatory management of complications (aspiration, respiratory failure, status epilepticus).

14. Other Species / Natural Disease

No naturally-occurring COXPD51-equivalent disease has been reported in companion animals or wildlife (no OMIA entry). Conserved orthologs exist across metazoans and yeast (mouse *Ptcd3* GeneID 69956; rat 500199; zebrafish *ptcd3* 553325; yeast *RSM22*), reflecting deep evolutionary conservation of the mitochondrial translation machinery. No zoonotic dimension (non-infectious genetic disease).

15. Model Organisms

- **In vitro / cellular:** 143B osteosarcoma cells with siRNA knockdown ([PMID: 19427859](#)); patient-derived dermal fibroblasts with lentiviral complementation rescue ([PMID: 30607703](#)).
- **Mouse:** IMPC *Ptcd3* knockout — homozygous embryonic lethality (gene essentiality); heterozygous impaired pupillary reflex. **No conditional/tissue-specific or hypomorphic mouse model recapitulating COXPD51 has been published** — a major gap.
- **Phenotype recapitulation:** cellular models faithfully reproduce the biochemical defect (combined CI/CIV deficiency, translation defect) but not the whole-organism neurodegenerative phenotype; the constitutive KO mouse dies too early to model Leigh syndrome.

Limitations and Knowledge Gaps

1. **Extreme rarity (n = 4 confirmed patients).** All clinical, prognostic, and genotype–phenotype inferences rest on a handful of cases plus proxy data from broader Leigh-syndrome cohorts. Prevalence, incidence, sex ratio, penetrance, and expressivity are essentially unquantified.

2. **No dedicated animal disease model.** The constitutive *Ptcd3* knockout is embryonic-lethal, so a **conditional (e.g., neuron-specific) or hypomorphic knock-in** model is needed to study disease pathogenesis and test therapies in vivo. This is the single largest gap.
 3. **Prognosis is inferred, not measured.** COXPD51-specific survival and progression data do not exist; the Leigh-syndrome proxies ([PMID: 31967322](#); [PMID: 32000367](#)) may not perfectly capture PTCD3-specific outcomes.
 4. **Genotype–phenotype correlation is immature.** The role of the hypomorphic p.Pro640Ala allele in modulating severity is inferred from allele frequency and cell data, not from systematic patient-level correlation.
 5. **No therapeutic evidence.** No trial, no approved therapy, and no biomarker of treatment response are available for COXPD51 specifically.
 6. **Structural biology of mS39 in disease.** How specific missense variants (e.g., p.Thr301Ile, p.Pro640Ala) perturb mS39 folding, RNA binding, or mt-SSU assembly has not been resolved structurally.
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Proposed Follow-up Experiments / Actions

1. **Build a tractable animal model.** Generate a **conditional (neuron-specific, e.g., Nestin-Cre) or hypomorphic knock-in *Ptcd3* mouse** that survives past organogenesis, to reproduce the Leigh phenotype and provide a platform for preclinical therapy testing. Zebrafish *ptcd3* morphants/mutants offer a faster, complementary vertebrate model.
2. **Establish an international patient registry / GeneMatcher effort.** Aggregate additional PTCD3 cases to define natural history, prevalence, genotype–phenotype correlations, and biomarkers. Curate against MSeqDR/PMD-VR mitochondrial-disease registries ([PMID: 42510797](#)).
3. **Functionally classify the VUS burden.** Deploy a high-throughput saturation/complementation assay in *PTCD3*-null cells to reclassify the 129 ClinVar VUS (especially recurrent missense variants), improving diagnostic yield.
4. **Structural characterization.** Use cryo-EM of the human mt-SSU and AlphaFold-based modeling to map how p.Thr301Ile, p.Pro640Ala, and other missense variants affect mS39–rRNA contacts and subunit assembly.
5. **iPSC-derived neuronal models.** Differentiate patient (or CRISPR-engineered) iPSCs into cortical/dopaminergic neurons and cerebral organoids to model the tissue-specific energy

failure and screen candidate therapeutics (e.g., mitochondrial biogenesis inducers, PPR-stabilizing compounds).

6. **Empirical supportive-care consensus.** Given the absence of disease-specific therapy, adopt and document outcomes under existing mitochondrial-disease supportive-care frameworks (e.g., ERN EURO-NMD guidance, [PMID: 40273815](#)) to standardize management across the few known patients.

Conclusion

COXPD51 is a mechanistically transparent but clinically devastating ultra-rare mitochondrial disease. The causal chain — **biallelic *PTCD3* loss of function → mS39 deficiency → defective 28S mitoribosome assembly → impaired mitochondrial translation → combined complex I + IV deficiency → neuronal energy failure → infantile Leigh syndrome** — is supported by patient genetics, biochemical assays, complementation rescue, and gene-essentiality data from the embryonic-lethal mouse knockout. The principal unmet needs are a survivable in-vivo disease model, systematic natural-history data, functional reclassification of VUS, and any disease-modifying therapy. Until those exist, diagnosis rests on exome/genome sequencing with biochemical confirmation, and care remains entirely supportive.

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